LEVELS OF COGNITIVE DEMAND

Four ways mathematical tasks engage thinking

4 HIGHER-LEVEL DEMAND

DOING MATHEMATICS

- No familiar algorithm or predictable solution path is provided.
- Explore concepts, relationships, and task constraints.
- Draw on relevant knowledge; monitor and adjust reasoning.
- Requires considerable cognitive effort; uncertainty may arise.

3 HIGHER-LEVEL DEMAND

PROCEDURES WITH CONNECTIONS

- Use procedures to deepen understanding of mathematical ideas.
- Connect procedures to diagrams, symbols, and representations.
- Follow general methods grounded in underlying concepts.
- Think about why a procedure works, not just the steps.

2 LOWER-LEVEL DEMAND

PROCEDURES WITHOUT CONNECTIONS

- Carry out a specified or familiar step-by-step procedure.
- Steps and expected outcomes are clear, with little ambiguity.
- Emphasis is on correct answers rather than underlying meaning.
- Explanations, if any, describe the steps performed.

1 LOWER-LEVEL DEMAND

MEMORIZATION

- Recall or reproduce previously learned facts, rules, or formulas.
- The requested response is explicit and unambiguous.
- No procedure is used, or there is insufficient time to use one.
- Understanding underlying concepts is not required by the task.

IMPORTANT DISTINCTION

Cognitive demand concerns the kind of thinking a task requires, not simply its difficulty or the time it takes. Demand can change in implementation.

Adapted from Stein, Grover, & Henningsen (1996), American Educational Research Journal, 33(2), 455–488.
Poster design and adaptation © 2026 Luis Miguel Fernández, Ph.D.